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MATHEMATICS
Paper 1
June/July.2026
$2\frac{1}{4}$ hours

BOARD OF GOVERNORS EXAMINATION CONSULTS

Uganda Certificate of Education

PRE-MOCK ASSESSMENT

MATHEMATICS

Paper 1

2 hours 15 minutes

INSTRUCTIONS TO CANDIDATES

*This examination paper has **two** Sections; **A** and **B**. it consists of **six** items.*

*Section **A** has **two compulsory** items.*

*Section **B** has **two** parts **I** and **II**. Respond to **one** item from each part.*

*Respond to **four** examination items in all.*

*Any additional item(s) responded to will **not** be scored.*

***All** responses **must** be written in the answer booklet(s) provided.*

Graph Paper is provided.

Silent, non-programmable scientific calculators and mathematical tables with a list of formulae may be used.

Turn Over

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SECTION A

Attempt **both** *items in this section*

ITEM 1

A certain company is to deploy its workers in three districts Kamuli, Jinja and Iganga. Jinja is south from Kamuli and Iganga is east from Jinja. To avoid errors in decimals, the distances from Kamuli to Jinja and Jinja to Iganga are given as $\sqrt{40{,}000}\ km$ and $\sqrt{22{,}500}\ km$ respectively. They have enough fuel to drive from Kamuli to Jinja then to Iganga, but they will need to re-fill the bus at a petrol station located at Iganga so that they leave with enough fuel to drive them back directly to Kamuli from Iganga. The bus they are to use consumes 0.3 litres of fuel per km.

In district Kamuli, Jinja and Iganga, they are to deploy groups of 12, 56 and 15 workers respectively. The company manager wants to know the least common number of products that can be given to each group, such that they are divisible amongst all the workers. He also needs to keep a record of the products each worker will have to sale in the respective districts.

A survey was carried out in Soroti district on a sample of 400 people. It was discovered that 15% of the people disliked the product. Those who had no knowledge about the product were $1\frac{1}{3}$ of those who disliked the product. 0.2 of the remainder already had the product and the rest ordered the product. The manager is to deploy other workers in the district if the percentage of those who ordered the product was 50%.

Tasks

a) How many liters of fuel will they have to re-fill the bus with?
b) What is the least common number of products that the manager will give to each group of workers?
c) How many products will each worker have to sell in the three districts?
d) Will the manager deploy more workers in Soroti district? (Justify your answer).

ITEM 2

Agriculture students in a certain school plan to produce and pack liquid soap and hand sanitizers for a community health exhibition. They plan to sell a bottle of liquid soap at shs.3,000 and a bottle of sanitizer at shs.5,000. To optimize their project; they have intend to prepare at most 1,200 mL of chemicals, where each bottle of liquid soap requires 20 mL of chemicals while each bottle of sanitizer requires 30 mL; due to bottle container availability, the number of sanitizer bottles produced must not exceed three

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times the number of liquid soap bottles produced; and the total number of bottles produced must be at least 25 bottles to qualify for provision of a space for exhibition.

On the exhibition day, the students expect to sell chemicals worth shs.15,000 per hour, and they intend to operate for 6 hours only. They incurred a total production cost of shs.35,000 and would wish to have an expected profit of at least 50% from the sales.

Tasks

(a) Formulate inequalities for the constraints, draw the feasible region, and recommend the ideal number of liquid soap and sanitizer bottles to maximize revenue.

(b) Based on their expected sales per hour and cost of production, determine whether their expected profit is at least 50%.

SECTION B
PART I

Respond to **only one** *item in this part*

ITEM 3

In a certain high school, learners with a good background in science were to be selected to participate in an Inter-School Science Challenge (ISSC). Those who perform well in at least two of the subjects; Physics, Chemistry and Biology are to represent the school in quiz competitions.

An analysis of learners' performance records revealed that; 9 learners do not perform well in any of the three science subjects. The number of learners who perform well in Biology only is three more than those who perform well in Chemistry only. 7 learners perform well in both Chemistry and Biology. 8 learners perform well in both Physics and Chemistry. The number of learners who perform well in Physics and Biology but not Chemistry is one less than the number of learners who perform well in all three subjects. The number of learners who perform well in Physics only is one more than those who perform well in Chemistry only. A total of 24 learners perform well in Physics. A total of 21 learners perform well in Chemistry.

The school would like to analyse the learners' performance in order to determine the most suitable candidates for the competition.

The school would also like to obtain the percentage of learners that are good at only one of the subjects which will be used by the headteacher to decide whether to carry out an appraisal on science teachers or not if its greater than 60%.

Tasks

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a) Organize the data above in a relevant presentation while determining all the un-given data.
b) How many learners will represent the school in the quiz competitions?
c) Using calculations, justify whether the headteacher carry out an appraisal on science teachers or not?

ITEM 4

The table below shows the sales in millions made by a certain company over a given period of time;

54	**67**	**45**	**62**	**56**	**32**	**42**	**45**	**54**	**46**
33	**71**	**58**	**70**	**49**	**68**	**56**	**34**	**33**	**43**
61	**41**	**57**	**67**	**31**	**72**	**48**	**73**	**66**	**78**
49	**35**	**44**	**34**	**66**	**59**	**52**	**63**	**49**	**69**
69	**71**	**72**	**42**	**51**	**50**	**62**	**40**	**69**	**40**
34	**64**	**41**	**60**	**33**	**53**	**35**	**79**	**54**	**79**

The owner of the company wants to analyze the data to make some conclusions and decisions;

- The average sales are to show him that the business has improved if they are at least fifty million.
- He will increase on adverts if the percentage number of times within which the sales were 49.5 million and below was at most 50%.

Tasks

a) Group the data into five classes and display it using a relevant statistical presentation.
b) Did the business performance improve? Justify your response.
c) Will the owner of the company increase on adverts? Justify your answer.

PART TWO

*Respond to **only one** item in this part*

ITEM 5

A prominent businessman wants to invest UGX.60,000,000 in the hard ware business in the one of the cities in Uganda. He wants to get a loan which is 30% of the investment. He approached a micro-finance officer who lends at 10% per annum simple interest rate for 4 years. The lender conditioned him to pay the principal and interest in equal monthly instalments. The businessman wants to hire a hardware manager and pay him UGX.850,000 which includes transport allowances of UGX.50,000 and UGX.55,000 respectively at the end every month.

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The manager will also be given a family allowance for only two of the four children aged 4, 8 and 14 years according.
A child below 5 years gets UGX.20,000, a child be ween 5 and 10 years UGX.10,000 and one above 12 years gets UGX.30,000.
His income is subjected to taxation according to the tax structure below.

Taxable income (UGX.)	**Tax rate (%)**
]1-160,000	free
160,001-350,000	10
Above 350,000	15

The manager has been earning a net pay of UGX.600,000 from his previous job. He is finding it difficult to decide on whether to take up the new job offer or not.

Tasks

(a) What is the total amount of money the business man will pay to the lender monthly?
(b) How much will be deducted monthly by the government as a tax?
(c) Do you advise the manager to take up this new job or not?

ITEM 6

John is planning to design a covered store for his animals and needs the roof raised to a single top point above a rectangular base that measures 4m by 6m and each slant edge from a base (floor) corner up to the top point will be 13m. Before he buys materials to be used, he wants to estimate how much space the store will provide inside and he also wants to know how steep one roof face is to the rectangular base (floor) level so that he can choose suitable roofing sheets and support.
For the design to be accepted, the height of the store must not be less than 10m or more than 12m, with each slanting surface inclined with at least 35^{0} to the base.

Tasks

Help the John to;
(a) Come up with an illustration that meets his requirements.
(b) Find how much space the store will provide inside.
(c) Determine how steep one roof face is to the rectangular base (floor) level.
(d) Know whether the design will be accepted or not?

END

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